

ADSORPTION OF LIPID BILAYERS TO MONOLAYERS - A NEW TRIPLE LAYER SYSTEM FOR STUDYING MEMBRANE PROTEINS

Christian SCHWIEGER¹, Isabel BERG¹, Jasmin PENKER¹, Vanessa
JERSCHABECK¹, Luisa VOIGT¹, Felix GOETZE¹, Dariush HINDERBERGER¹

¹ Martin Luther University Halle-Wittenberg, Institute of chemistry, Halle (Saale), Germany
(Christian.schwieger@chemie.uni-halle.de)

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ABSTRACT

Lipid monolayers can be used as very simple model system for one membrane leaflet. They have the advantages of being perfectly oriented, completely hydrated, and accessible through the aqueous phase. Thus, they are well suited for interaction studies with membrane binding compound, such as proteins or peptides.

However, no membrane spanning molecules can be studied in this system until now. Therefore, we are aiming to design a new membrane model system that makes the many advantages of established monolayer methods accessible for more complex samples.

We are exploring methods to assemble lipid bilayers underneath lipid monolayers at the air-water interface. The bilayers will be oriented, allowing to deduce molecular orientations from experiments, while still being in a well-hydrated and natural environment. We will present successful attempts in constructing such a model system, i.e., the adsorption of different lipid nanodiscs or vesicles to lipid monolayer at the air-water interface.

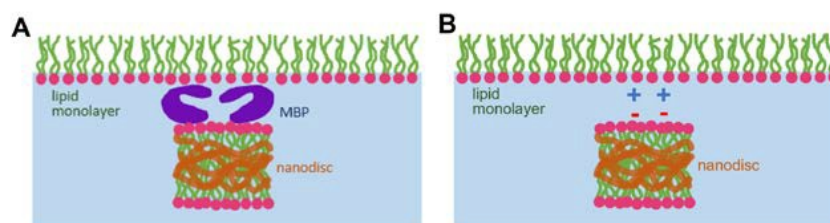


Fig. Constructing lipid triple layers at the air-water interface by adsorption of lipid nanodiscs mediated by MBP (myelin basic protein) (A) or electrostatically driven (B)